Reg. No.															

B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, DECEMBER 2025

Fourth, Seventh Semester

21CSE255T - COMPUTER GRAPHICS AND ANIMATION

(For the candidates admitted during the academic year 2021 - 2022 to 2024 - 2025)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks)

Answer **all** Questions

	Marks	CO	BL
1. The original coordinates of the point in polar coordinates are (A) X'=r cos (Φ +Θ) and Y'=r cos (Φ +Θ) (B) X'=r cos (Φ +Θ) and Y'=r sin (Φ +Θ) (C) X'=r cos (Φ -Θ) and Y'=r cos (Φ -Θ) (D) X'=rcos (Φ +Θ) and Y'=rsin (Φ -Θ)	1	1	2
2. _______ controls the basic display properties of output primitives (A) Attribute parameter (B) setpixel() (C) getpixel() (D) parameter	1	1	1
3. The window opened on the raster graphics screen in which the image will be displayed is called _____________ (A) World co-ordinate system (B) Screen co-ordinate system (C) World Window (D) Interface Window	1	1	1
4. In LCD, the refresh rate of the screen is (A) 60 frames/sec (B) 80 frames/sec (C) 30 frames/sec (D) 100 frames/sec	1	1	1
5. How many phosphor color dots at each pixel position in a shadow mask CRT? (A) 1 (B) 7 (C) 2 (D) 3	1	2	2
6. The scale factor of viewport transformation for x co-ordinate is ________ (A) Sx = (svmax – svmin)/ (swmax – swmin) (B) Sx = (svmax – svmin)/ (swmax + swmin) (C) Sx = (svmin – svmax)/ (swmax – swmin) (D) Sx = (svmax + svmin)/ (swmax – swmin)	1	2	1
7. In Liang–Barsky algorithm, a line is represented as: (A) y=mx+c (B) x=x1+t(x2−x1), y=y1+t(y2−y1) (C) Region Out codes (D) Homogeneous Coordinates	1	2	2
8. In the Cohen–Sutherland line clipping algorithm, the clipping window is defined as: xmin=2, xmax=6, ymin=2, ymax=6. If a point P(1,7)P(1,7)P(1,7) lies outside this window, what will be its region code? (A) 0100 (B) 1010 (C) 0010 (D) 1001	1	2	2.

9. What is the primary characteristic of perspective projection in OpenGL? 1 3 1

(A) Objects are projected onto the view plane without any distortion
(B) Objects appear larger when they are closer to the viewer
(C) Objects appear distorted if they are not perpendicular to the viewing direction
(D) Objects maintain their size regardless of their distance from the viewer

10. In 3D clipping, what does the term "view volume" refer to? 1 3 2

(A) The space within which objects are visible to the viewer
(B) The total number of pixels on the screen
(C) The area outside the viewing window
(D) The volume of space occupied by the camera

11. Consider point P(6,6,−4) if it is reflected over the XZ plane, then what are the final coordinates? 1 3 2

(A) (6, -6, -4)
(B) (-6, -6, -4)
(C) (6, 6, -4)
(D) (-6, 6, -4)

12. In 3D computer graphics, what effect does applying a shear transformation have on an object? 1 3 1

(A) It changes the object's size uniformly along all axes
(B) It rotates the object around a specified axis
(C) It skews the object's shape along one or more axes
(D) It translates the object along a specified direction in 3D space

13. Which of the following is not a hidden surface removal algorithm 1 4 1

(A) Sutherland Cohen algorithm
(B) Warnock algorithm
(C) Buffering algorithm
(D) Binary space partitioning algorithm

14. Choose the hidden surface removal algorithm from the below 1 4 1

i. Depth comparison, Z-buffer, back-face removal
ii Scan line algorithm, priority algorithm
iii. BSP method, area subdivision method

(A) i,ii & iii
(B) ii & iii
(C) i & iii
(D) i & ii

15. The scan line coherence algorithm was developed by 1 4 1

(A) Wylie
(B) Evans
(C) Cat mull
(D) Wylie & Evans

16. In which year Z- buffer algorithm is described? 1 4 1

(A) 1995
(B) 1974
(C) 1945
(D) 1981

17. In HLS color model, When S is decreased, the colors move toward _______ 1 5 1

(A) Gray
(B) Blue
(C) White
(D) Black

18. Which one is not a standard motion of a camera movement. 1 5 1

(A) Zooming
(B) Tilting
(C) Rendering
(D) Panning

19. ________ is a specialized animation languages designed simply to generate the in-betweens from the user-specified key frames. 1 5 1

(A) Java (B) Python
(C) C++ (D) Key-frame System

20. The National Television System Committee (NTSC) color model for forming the composite video signal is the ________ 1 5 1

(A) XYZ Model (B) HLS Model
(C) YIQ Model (D) HSV Model

PART - B (5 × 8 = 40 Marks)

Answer all Questions

Marks CO BL

21. a) Suppose you are developing a graphics application, and you need to draw a line on a pixel grid using Bresenham's line drawing algorithm. The line should start from point (13, 10) and end at point (15, 20). Write the Algorithm of Bresenham's Line drawing. Walk through the steps of Bresenham's algorithm, calculating the decision parameter at each step and determining the pixels to be plotted. Provide the coordinates of each pixel along the line. 8 1 3

(OR)

b) Imagine you are tasked with drawing a horizontal line from point (5, 6) to point (8, 12) using the DDA algorithm. Calculate the increments and determine the pixels along this horizontal line. Write the steps of DDA line drawing algorithm for drawing a horizontal and vertical line. 8 1 3

22. a) A polygon with vertices: P_1 (10, 10), P_2 (80, 10), P_3 (80, 50), P_4 (10, 50) is to be clipped against a rectangular clipping window defined by:
$x\_min = 20$, $y\_min = 20$, $x\_max = 70$, $y\_max = 40$. Apply the Sutherland–Hodgman Polygon Clipping Algorithm and determine the vertices of the clipped polygon. 8 2 3

(OR)

b) A CAD allows the user to reflect the shape across the specified axis. Given a triangle with vertices A(3,4), B (6,7) and C(8,3). Apply 2D reflection across the X-axis, Y-axis and determine the new coordinates respectively. Also draw the corresponding diagrams 8 2 3

23. a) Describe the role of 3D scaling in computer graphics, and outline how it impacts objects without directly discussing formulas. If a point P(2, 3, 4) undergoes uniform scaling by a factor of 2, what will be its new coordinates after scaling? 8 3 3

(OR)

b) Explain the significance of 3D reflection in computer graphics and its real-world applications. Furthermore, if a point P(2, 3, 4) undergoes a reflection operation about the xy-plane, what will be the resulting coordinates of the point after reflection? 8 3 3

24. a) outline the differentiate point light source and distributed light source. 8 4 2

(OR)

b) Illustrate diffuse reflection and specular reflection. 8 4 2

25. a) Summarize the twelve principles of animation, and why are they considered essential guidelines for creating compelling and believable motion in animated sequences? 8 5 2

(OR)

b) Identify some practical applications of contrast stretching and image enhancement techniques in computer graphics, such as in rendering, digital art, or multimedia presentations? 8 5 2

PART - C (1 × 15 = 15 Marks)

Answer **any 1** Question

	Marks	CO	BL
26. You are a filmmaker working on a new animated movie set in a fantastical world filled with mythical creatures and breathtaking landscapes. As part of the animation process, you need to utilize perspective projection techniques to create dynamic and visually stunning scenes that transport viewers into the heart of the story. Your task is to understand the principles of perspective projection and explore how different types of perspective projection can be applied to enhance the storytelling and visual appeal of the movie.	15	3	2
27. As a VR game developer for an advanced gaming console, you're tasked with optimizing rendering for immersive experiences. To overcome challenges, you plan to utilize ray casting method. Its advantage lies in efficiently simulating realistic lighting and shadows, enhancing visual quality. Describe your strategy for utilizing the ray casting method to overcome rendering challenges in the VR application for the advanced gaming console. Highlight the method's advantages in this context and detail any optimizations or techniques you plan to implement to ensure both smooth performance and exceptional visual quality.	15	4	2

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